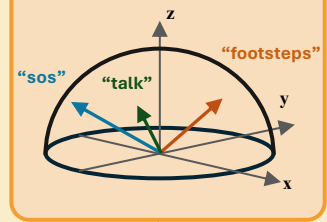
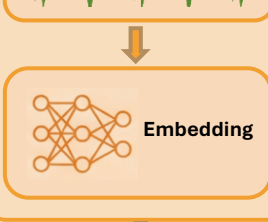


Perception Layer Multimodal Sensing & State Estimation

Embed-ACCDQA
SELD



separated sound



Audio Entities

RGB-D

YOLOv11 Detection



Depth



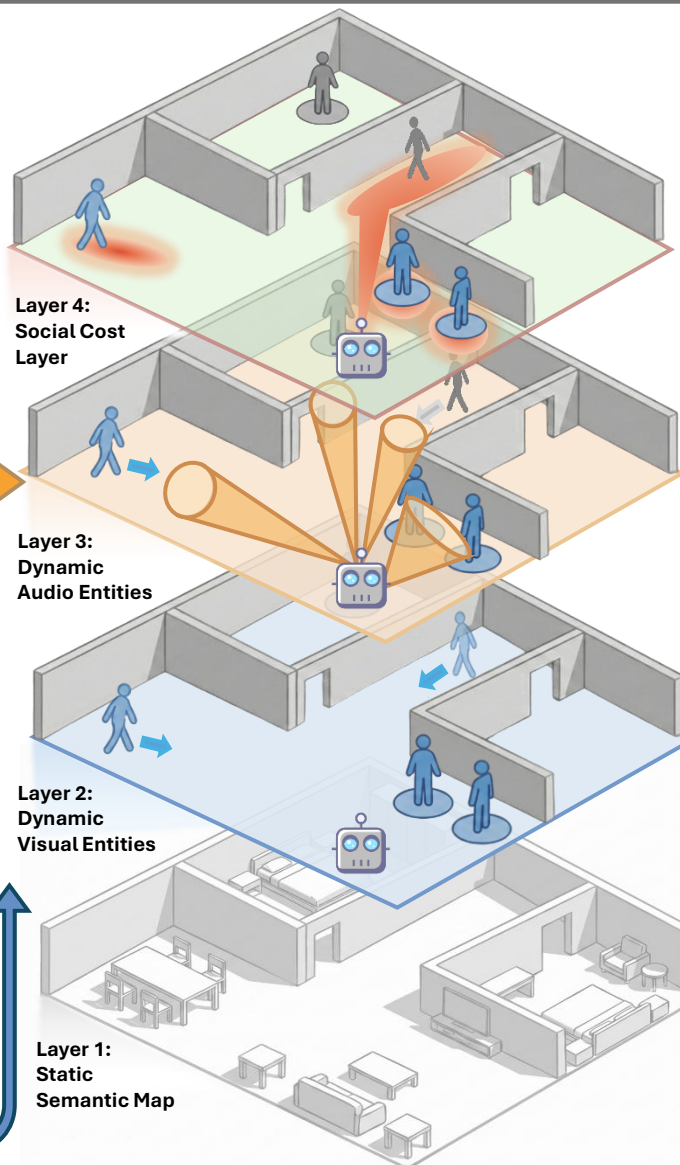
Visual Perception



Embedding

Visual Entities

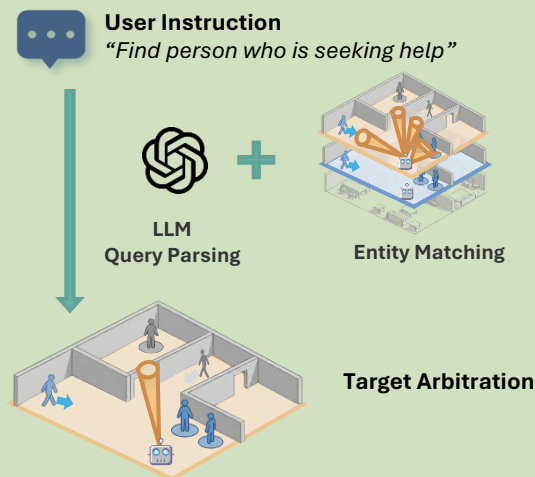
SAVMap: Social-Aware Audio-Visual Mapping



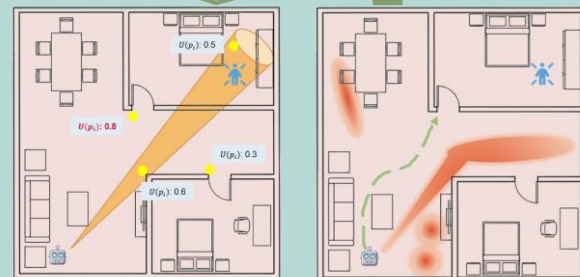
Social Costmap



Planning Layer: Dual-Mode Navigation & Active Exploration



Active Audio-Visual Search



Social-Aware Navigation